

The Vacuum Catastrophe and the Granularity of Hilbert Space

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Quantum field theory with a Planckian cutoff predicts a vacuum energy density of order the Planck density, $4.6 \times 10^{113} \text{ J m}^{-3}$, 123 orders of magnitude above what the observed cosmological constant allows. This paper resolves the discrepancy in two steps, within a framework in which every quantum state carries the same finite budget of L bits and gravity follows from that limit. First, the zero-point energy does not gravitate: a stress tensor proportional to the metric has no flux through any null surface, so it never enters the thermodynamic derivation of Einstein's equations, whatever its magnitude, and Λ survives as an integration constant that the horizon fixes. Second, the standard estimate's error is in the state count. The energy Λ carries inside the horizon is the horizon entropy times the horizon temperature, which gives $\rho_\Lambda = 3\pi\rho_{\text{Planck}}/2L^2$, with $L = 6.4 \times 10^{61}$ the number of Planck lengths around the horizon: one factor of L because the count is of a boundary and not of a volume, and one because a pixel's energy scale is E_P/L and not E_P . The acceleration scale of galactic rotation curves, $a_0 = (1.20 \pm 0.24) \times 10^{-10} \text{ m s}^{-2}$, fixes L independently at $(4.6 \pm 0.9) \times 10^{61}$ and gives $\rho_\Lambda = (1.0 \pm 0.4) \times 10^{-9} \text{ J m}^{-3}$, against the observed 5.2×10^{-10} ; in the measured variable a_0 the two agree at 1.4σ , the residual being the order-unity coefficient the framework has yet to derive. The same L caps a state spread across its Hilbert space at 205 qubits, a ceiling whose measurement would be a third determination with no astronomy in it.

Keywords: Cosmological constant; vacuum energy; emergent gravity; holographic principle; discrete Hilbert space; rational quantum mechanics

I. THE PROBLEM

Quantum field theory assigns every mode of every field a zero-point energy $\frac{1}{2}\hbar\omega$. Summed to a Planckian cutoff, this gives a vacuum energy density of order the Planck density,

$$\rho_{\text{vac}} \sim \frac{c^7}{\hbar G^2} = 4.6 \times 10^{113} \text{ J m}^{-3}, \quad (1)$$

while the observed cosmological constant corresponds to

$$\rho_\Lambda = \frac{\Lambda c^4}{8\pi G} = 5.2 \times 10^{-10} \text{ J m}^{-3}, \quad (2)$$

a ratio of 8.8×10^{122} ; in Planck units, $\Lambda \ell_P^2 = 2.85 \times 10^{-122}$ [1, 2]. Adler, Casey, and Jacob named this the vacuum catastrophe [3], after the ultraviolet catastrophe it resembles. The cutoff is not the issue. Electroweak symmetry breaking and the QCD condensate contribute at $(100 \text{ GeV})^4$ and $(0.3 \text{ GeV})^4$, some 10^{55} and 10^{45} times the observed value, and no symmetry protects their sum [2]. General relativity requires every form of energy to gravitate [4], so either the vacuum energy is canceled to 123 decimal places or it is not a source of curvature.

This paper takes the second route, within the framework of Ref. [5], in which Hilbert space is granular with a single universal integer L and gravity is derived from that granularity. A uniform vacuum energy is not a source, because a stress tensor proportional to the metric has no flux through a null surface (Sec. III); this removes the

$10^{113} \text{ J m}^{-3}$ but leaves Λ as an unexplained integration constant. The scale of what remains is set by the number of degrees of freedom a horizon volume contains, L^2/π and not $(R/\ell_P)^3$, and by their energy scale, E_P/L and not E_P ; those two factors of L are the catastrophe (Sec. IV). Section V fixes L from galaxies, and Sec. VI says how the laboratory will.

II. THE FRAMEWORK

Palmer's Rational Quantum Mechanics [6, 7] keeps the Schrödinger equation and the Born rule but admits only those states, and only those measurement bases, for which a qubit $|\psi\rangle = \cos\frac{\theta}{2}|1\rangle + e^{i\phi}\sin\frac{\theta}{2}|{-1}\rangle$ has

$$\cos^2\frac{\theta}{2} = \frac{m}{L}, \quad \frac{\phi}{2\pi} = \frac{n}{L}, \quad (3)$$

with $m \in \{0, \dots, L\}$, $n \in \{0, \dots, L-1\}$, and L a large integer. Such a state is a string of L bits: the fraction of +1 bits is $\cos^2(\theta/2)$, and rotating the string by n places is the phase [7]. In Palmer's version L is a property of each qubit and ranges from 10^{64} to 10^{109} for current qubit technologies [7]. Reference [5] takes L to be one universal integer for all systems (Axiom 1) and identifies one step of quantum phase, $2\pi/L$, with one Planck length on a great circle of the de Sitter horizon (Axiom 2). That horizon is the sphere of radius $R_\Lambda = \sqrt{3/\Lambda} = 1.66 \times 10^{26} \text{ m}$ from beyond which no light will ever reach an observer once the expansion is driven by Λ alone. Axiom 2 reads

$$\ell_P = \frac{2\pi R_\Lambda}{L}. \quad (4)$$

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The de Sitter radius, rather than the present Hubble radius, is the length in Axiom 2 because L is a constant and R_Λ is the only cosmological length that does not change with time; that the present Hubble radius lies within a factor of 1.2 of it, the coincidence problem, is not addressed here. From the two axioms, Verlinde's entropic argument [8] returns $G = c^3 \ell_P^2 / \hbar$ and Jacobson's [9] returns Einstein's equations. The Gibbons–Hawking temperature of the horizon [10] is $k_B T_{\text{dS}} = \hbar c / 2\pi R_\Lambda = E_P / L$, with $E_P = \hbar c / \ell_P$ the Planck energy. And, up to an order-unity coefficient discussed in Sec. V, $a_0 = c^2 / L \ell_P$ is the acceleration below which galactic rotation curves flatten. Reference [11] restates the two axioms as one postulate: a ring of L Planck cells, the register, on which every quantum state is a pattern and which closes on a great circle of the horizon. Reference [12] adds a postulate on the address space of momenta, extends the ring to the whole sphere, and obtains the three dimensions of space by counting. The two quantities used below are the pixel count of the horizon and the energy scale of one pixel.

III. WHY THE ZERO-POINT ENERGY IS NOT A SOURCE

The framework derives Einstein's equations by Jacobson's argument [9]. The Clausius relation $\delta Q = T dS$ is imposed on every local Rindler horizon, the horizon seen by a uniformly accelerating observer, with T the Unruh temperature that observer measures in the vacuum, dS a quarter of the change in horizon area in Planck units, and δQ the energy flux through the horizon,

$$\delta Q \propto \int \lambda T_{\mu\nu} k^\mu k^\nu d\lambda dA, \quad (5)$$

where k^μ is the null generator of the horizon, the light ray that sweeps it out, and λ its affine parameter. A stress tensor proportional to the metric has no such flux, since $g_{\mu\nu} k^\mu k^\nu = 0$ for null k^μ [13]. The zero-point energy, $T_{\mu\nu} = -\rho_{\text{vac}} g_{\mu\nu}$, therefore never enters the derivation, whatever ρ_{vac} may be. Nor do the electroweak and QCD condensates, which are Lorentz invariant and of the same form. Lorentz invariance is what fixes that form: a cutoff on three-momentum breaks it and gives the zero-point sum a radiation-like pressure, but that pressure is an artifact of the regulator, and any covariant regularization returns $T_{\mu\nu} \propto g_{\mu\nu}$, at a magnitude set by the particle masses rather than by the cutoff [2]. The argument does not care which of the two sets the magnitude, since the term drops out whatever its size. What Jacobson's argument returns is

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}, \quad (6)$$

with Λ an integration constant [9, 14]; that the Clausius derivation yields the trace-free, unimodular form of gravity, in which the vacuum energy does not couple, has been shown in detail [15, 16]. Nothing in Eq. (6) prevents that

constant from being set equal to $8\pi G \rho_{\text{vac}} / c^4$, which would put the zero-point energy back by hand. What prevents it here is Axiom 2, which fixes Λ from the horizon and not from the field content. The same holds at the Newtonian level in Verlinde's derivation [8]: the force comes from the change in screen entropy as a mass is displaced, and a background that is the same on both sides of the screen produces none.

Decoupling and the integration constant are two of the three conditions Padmanabhan and Padmanabhan [17] identify for a solution of the cosmological constant problem: the field equations are unchanged by $\mathcal{L}_{\text{matter}} \rightarrow \mathcal{L}_{\text{matter}} + \text{const}$, and Λ enters solutions as an integration constant rather than as a parameter in the action; the third, a principle that fixes its value, is Axiom 2, which fixes Λ in terms of an integer that is measured. The first two hold in unimodular gravity, where the trace-free field equations are blind to a constant shift of $T_{\mu\nu}$ [18, 19]. Here they are not adopted; they are how gravity is derived.

What drops out is exactly the part of the energy density that is constant throughout spacetime, and the reason is the conservation step. The null flux vanishes for $\rho(x)g_{\mu\nu}$ with any $\rho(x)$, but Jacobson's derivation also takes the total stress tensor to be conserved, and the divergence of Eq. (6) with the Bianchi identity then admits only a constant: a vacuum energy that varies in space or in time is not conserved by itself, re-enters through the conservation of the total, and gravitates, as in unimodular gravity. Vacuum *differences* therefore weigh, and the derivation predicts as much. The vacuum-loop correction to the electrostatic energy of a nucleus, the same physics as the Lamb shift, is a shift in binding energy relative to the surrounding vacuum, localized and different for aluminum and platinum. Free-fall tests of the equivalence principle confirm that the correction weighs, to one part in 10^6 [20, 21] and to about one part in 10^9 with the MICROSCOPE bound of 3×10^{-15} on the Eötvös parameter of titanium and platinum [22]. So does the proton, most of whose mass is QCD binding energy: a localized excitation above the condensate, not the condensate itself. An integration constant is a boundary condition, and the boundary of a universe with $\Lambda > 0$ is its asymptotic horizon, after every transition. The vacuum energy before the electroweak and QCD transitions exceeded the constant that remains by $(100 \text{ GeV})^4$ and $(0.3 \text{ GeV})^4$, and those excesses, being differences from that constant, gravitated then, as vacuum energy before each transition and as radiation after it, exactly as in standard cosmology; what the horizon fixes is the constant that remains.

Equation (6) settles half of the problem. The zero-point energy, at the Planck density of Eq. (1) or at any other, is a constant multiple of the metric, has no flux through any null surface, and never enters the field equations. Whatever quantum field theory says the vacuum weighs, gravity does not respond to it.

The other half is Λ . It survives in Eq. (6) as an integration constant: the equation does not fix its value, and something outside the equation must. In the present

framework the horizon does. Axiom 2 ties the Planck length to the horizon through L , and with $\Lambda = 3/R_\Lambda^2$ this reads

$$\Lambda \ell_P^2 = \frac{12\pi^2}{L^2} = 2.85 \times 10^{-122}, \quad (7)$$

or, equivalently, $G\Lambda = 12\pi^2 c^3/\hbar L^2$: the strength of gravity and the acceleration of the universe are one constant, with L the exchange rate between them [5]. Because the horizon is large in Planck lengths, $L = 6.4 \times 10^{61}$ of them around, Λ is small in Planck units, and the question why Λ is small becomes the question why L is 6.4×10^{61} , a number the framework does not derive but measures, in Sec. V and prospectively in Sec. VI. The value is also stable. In unimodular gravity the observable is $\Lambda_0 + 8\pi G\rho_{\text{vac}}/c^4$ and inherits the sensitivity of ρ_{vac} to every loop, so the integration constant must be retuned against each [23]. Here the source in the Clausius relation is $T_{\mu\nu}k^\mu k^\nu$, so two stress tensors that differ by a constant multiple of $g_{\mu\nu}$ are one source: the constant part of ρ_{vac} is not a quantity the theory contains at any loop order, and what it does not contain cannot be tuned. The constant Eq. (6) carries is not $\Lambda_0 + 8\pi G\rho_{\text{vac}}/c^4$ but $12\pi^2/L^2\ell_P^2$, fixed by an integer, and a loop correction, which shifts $T_{\mu\nu}$ by a multiple of $g_{\mu\nu}$, is not a change of L . What remains to be said here is what the energy of Λ physically is, and why it comes to $5 \times 10^{-10} \text{ J m}^{-3}$ rather than the 10^{113} of the zero-point sum. The next section answers by counting the degrees of freedom on the horizon.

IV. THE STATE COUNT

The continuum estimates the energy of the vacuum from the bulk. A region of radius R is assigned $(R/\ell_P)^3$ zero-point modes at the Planck energy apiece, and, spread over the volume $\sim R^3$, these give ρ_{Planck} for any R ; that is Eq. (1). Section III shows that none of that energy gravitates. The constant Λ is not a source but a term in Eq. (6); de Sitter thermodynamics assigns it an energy, and the framework locates that energy on the horizon. The derivation takes five steps.

Step 1. Boundary count. Axiom 2 gives $R_\Lambda = L\ell_P/2\pi$. The horizon is a two-sphere, because space has three dimensions, which Ref. [12] obtains from the same L by counting. Its budget of degrees of freedom is its pixel count,

$$N_{\text{pix}} = \frac{4\pi R_\Lambda^2}{\ell_P^2} = \frac{L^2}{\pi} = 1.32 \times 10^{123}, \quad (8)$$

an area rather than a volume. Counted in the register's own cells, which have area ℓ_P^2/π on the horizon [12], the sphere has $N_{\text{cell}} = L^2$ cells instead; the two counts differ by a factor of π , and the entropy, $S_{\text{BH}}/k_B = N_{\text{pix}}/4 = N_{\text{cell}}/4\pi = L^2/4\pi$, is the same in either. The Planck pixel is the unit of area in the Bekenstein–Hawking entropy and is the one used here.

Step 2. Energy scale of a pixel. The Gibbons–Hawking temperature [10] of the horizon, $k_B T_{\text{dS}} = \hbar c/2\pi R_\Lambda$, with $2\pi R_\Lambda = L\ell_P$ from Axiom 2, sets the energy scale of one pixel,

$$\varepsilon = k_B T_{\text{dS}} = \frac{\hbar c}{L\ell_P} = \frac{E_P}{L} = 3.0 \times 10^{-53} \text{ J}, \quad (9)$$

the Planck energy divided by L , rather than the Planck energy; it is the register's energy quantum $\varepsilon_0 = \hbar c/R_\Lambda$ [11] divided by 2π .

Step 3. Total energy. The Bekenstein–Hawking entropy of the horizon, of area $A = 4\pi R_\Lambda^2$, is $S_{\text{BH}} = k_B A/4\ell_P^2 = \frac{1}{4}N_{\text{pix}}k_B$, and entropy times temperature is

$$E = S_{\text{BH}} T_{\text{dS}} = \frac{1}{4}N_{\text{pix}}\varepsilon = 1.00 \times 10^{70} \text{ J}. \quad (10)$$

This is exactly the energy Λ carries inside the horizon, $\rho_\Lambda V$ with $V = \frac{4}{3}\pi R_\Lambda^3$ the areal volume, an identity of de Sitter thermodynamics [24]. Equivalently, since for $p = -\rho$ the Tolman–Komar mass $\int(\rho + 3p)dV$, what focuses geodesics in general relativity, is $2\rho_\Lambda V$ in magnitude over the same V , the lapse of the static patch canceling its proper-volume factor,

$$2\rho_\Lambda V = \frac{1}{2}N_{\text{pix}}\varepsilon, \quad (11)$$

one half-quantum $\frac{1}{2}k_B T_{\text{dS}}$ per pixel, which is Padmanabhan's holographic equipartition [17, 25, 26] on the de Sitter horizon.

Step 4. Energy density. Dividing by $V = \frac{4}{3}\pi(L\ell_P/2\pi)^3$,

$$\rho_\Lambda = \frac{E}{V} = \frac{3\pi}{2} \frac{E_P}{L^2\ell_P^3}. \quad (12)$$

Step 5. In Planck units. With $\rho_{\text{Planck}} = E_P/\ell_P^3 = c^7/\hbar G^2 = 4.63 \times 10^{113} \text{ J m}^{-3}$,

$$\boxed{\frac{\rho_\Lambda}{\rho_{\text{Planck}}} = \frac{3\pi}{2L^2} = 1.13 \times 10^{-123}}, \quad (13)$$

and $\rho_\Lambda = 5.25 \times 10^{-10} \text{ J m}^{-3}$. As an energy scale this is $\rho_\Lambda^{1/4} = (3\pi/2)^{1/4} E_P/\sqrt{L} = 2.2 \text{ meV}$, the millielectronvolt of dark energy written in the framework's constants. One factor of L is lost to the dimensionality of the count, Step 1, and one to the energy scale of a pixel, Step 2.

Steps 3 and 4 are identities of de Sitter thermodynamics and hold in general relativity for any Λ : the entropy of a de Sitter horizon times its temperature, spread over its interior, is the Λ that produces the horizon [24]. Steps 1, 2, and 5 use Axiom 2, and, given a value of L from any source, they predict the pixel count, the energy scale of a pixel, and the vacuum energy density. What Axiom 1 adds is that the integer L , which fixes the number of states inside the horizon, $e^{L^2/4\pi}$ by Step 1, is the same L that bounds every quantum state [11]. The horizon count is then not merely an area in Planck units but the information capacity of the universe: Bousso's N -bound [27] and Banks' proposal that a positive Λ and a

finite Hilbert space go together [28], written in terms of one integer. That a finite-dimensional Hilbert space tames the vacuum energy has been argued by constructing field modes of finite dimension [29]; here the zero-point energy is not reduced but decoupled, and what is finite is the horizon's count. With L taken from the horizon, as here, Eq. (13) returns the observed density by construction; the test is Sec. V, where L comes from galaxies.

The exponent is robust; the coefficient is Axiom 2's. The value $L = 6.4 \times 10^{61}$ is read off a known length: one does not need Λ to know that a cosmological horizon is some 10^{26} m in radius, and the exponent does not depend on which horizon is meant. The Hubble radius gives $(R/\ell_P)^{-2} = 10^{-121.9}$, the de Sitter radius $10^{-122.0}$, and the particle horizon $10^{-122.9}$: a spread of one decade across lengths differing by a factor of three, against a problem of 123 decades. The scaling $\rho_\Lambda \sim \rho_{\text{Planck}}(\ell_P/R)^2$ has been reached before, from effective field theory [30], from holography with the Hubble radius [31] or the future event horizon [32] as the infrared cutoff, from mode counting [33, 34], and from vacuum fluctuations [35]. Equation (13) is the holographic form with its free coefficient fixed at one and its cutoff the constant R_Λ , so that $w = -1$ exactly, where an evolving event horizon gives $w_0 \approx -0.9$ [32]. Axiom 2 is itself a relation between the ultraviolet and infrared cutoffs, $\ell_P/R_\Lambda = 2\pi/L$, with its coefficient fixed. What the framework adds is that L is a universal integer with two determinations those derivations do not have, one made and one to come: the galactic acceleration scale (Sec. V), which tests the coefficient, and the qubit ceiling (Sec. VI), which tests the exponent.

The 123 orders of magnitude are therefore L^2 , the square of the number of Planck lengths around the horizon. The continuum's estimate uses $(R_\Lambda/\ell_P)^3 = (L/2\pi)^3$ bulk modes at E_P ; the energy that gravitates is that of L^2/π boundary pixels at the scale E_P/L ; and the ratio is L^2 , up to a numerical factor.

V. L FROM GALAXIES

Reference [5] fixes L a second time, from data that do not involve Λ . The acceleration scale of the radial acceleration relation, $a_0 = (1.20 \pm 0.02_{\text{stat}} \pm 0.24_{\text{sys}}) \times 10^{-10} \text{ m s}^{-2}$, is measured from galactic rotation curves [36], the systematic term coming from the stellar mass-to-light ratio. Then $a_0 = c^2/L\ell_P$ inverts to $L_{\text{gal}} = c^2/a_0\ell_P = (4.6 \pm 0.9) \times 10^{61}$, and, carried through Eqs. (7) and (13), this gives

$$\Lambda_{\text{pred}} = \frac{12\pi^2}{\ell_P^2 L_{\text{gal}}^2} = \frac{12\pi^2 a_0^2}{c^4} = (2.1 \pm 0.8) \times 10^{-52} \text{ m}^{-2}, \quad (14)$$

or, in the units of Eq. (1), a vacuum energy density of $(1.0 \pm 0.4) \times 10^{-9} \text{ J m}^{-3}$. The observed values are $1.09 \times 10^{-52} \text{ m}^{-2}$ [37] and $5.2 \times 10^{-10} \text{ J m}^{-3}$. Prediction and observation agree to within a factor of 1.9: in the

measured variable, the horizon value of L corresponds to $a_0 = 0.86 \times 10^{-10} \text{ m s}^{-2}$, 1.4σ from the measurement. The 123 decades of Eq. (1) are thus closed from rotation curves alone, a measurement unrelated to cosmological surveys. Equation (14) contains neither $\hbar$ nor G : what rotation curves test is the coefficient of Axiom 2, and the quantum content of the exponent, L^2 with ℓ_P , is tested only by the qubit ceiling of Sec. VI.

Milgrom noticed in 1983 that $a_0 \sim cH_0$ [38], proposed in 1999 that it arises from the Unruh temperature of de Sitter space [39], and has since written the coincidence as $2\pi a_0 \approx cH_0 \approx c^2\sqrt{\Lambda/3}$, leaving open which of the two a_0 is tied to [40]. Equation (14) turns the coincidence into a relation, $\Lambda \propto a_0^2/c^4$, tied to R_Λ rather than to H_0 , with a coefficient the framework fixes up to one order-unity factor. That factor is the coefficient γ in $a_0 = \gamma c^2/L\ell_P$, which Ref. [5] does not derive and Refs. [11, 12] leave open. What is new in Eq. (14) is therefore not the relation, which is Milgrom's, but that it is tied to the constant R_Λ , so that a_0 does not evolve, and that a_0 and Λ are both functions of one integer that the qubit ceiling measures with no astronomy. The coefficient is unsettled throughout emergent gravity. Verlinde's derivation gives $cH_0/6$, with H_0 the Hubble rate of the de Sitter space it is set in [41]: $1.1 \times 10^{-10} \text{ m s}^{-2}$ at the present Hubble rate and $c^2/6R_\Lambda = 0.90 \times 10^{-10} \text{ m s}^{-2}$ at the de Sitter radius, 5% from $\gamma = 1$. Fits of rotation curves within emergent gravity prefer a scale about 30% below the present-rate value [42, 43]. The 2π in Axiom 2 itself is not free: it is the ratio of circumference to radius of the ring of L cells that Ref. [11] places on a great circle, and Ref. [12] finds the horizon's independently defined pixel to hold π of those cells, a factor of order one and not a power of L . The measured a_0 gives $\gamma = 1.39 \pm 0.28$ with the systematic uncertainty of Ref. [36], and 1.38 ± 0.11 with the tighter mass-to-light calibration of Ref. [44], $a_0 = (1.19 \pm 0.04_{\text{stat}} \pm 0.09_{\text{sys}}) \times 10^{-10} \text{ m s}^{-2}$. The framework's own scale, $\gamma = 1$, lies at 1.4σ on the first error and 3.3σ on the second; a coefficient half as large lies at 3.2σ on the first, and the bare surface gravity of the horizon, c^2/R_Λ , at 18σ . The remaining factor of 1.9 between Eq. (14) and the observed Λ is γ^2 , and a derivation of γ is what the framework owes.

VI. TEST

A state of N qubits spread across its Hilbert space has 2^N outcomes of nonzero probability. In the framework each probability is a frequency over L trials, hence at least $1/L$, and the probabilities sum to one; so $2^N \leq L$, and

$$N_{\text{max}} = \lfloor \log_2 L \rfloor = 205, \quad (15)$$

or 204 at the galactic value of L [11]. This is Palmer's $\log_2 L$ [7]; his bit count $2^{N+1} - 2 \leq NL$, which gives 212 and which Ref. [5] called exact, is a weaker necessary condition [11]. Beyond N_{max} , algorithms that spread

amplitude across the whole of Hilbert space lose their exponential advantage. Operationally, a random circuit on N fault-tolerant logical qubits followed by its inverse returns to its initial state with probability set by the gate fidelity in standard quantum mechanics, and here with probability only about $(L/2^N)^2$ once N exceeds 205 [11]. The ceiling is technology independent, the same for superconducting qubits as for trapped ions, which distinguishes it from every known decoherence mechanism. A measured $N_{\max}$ would fix L from the laboratory to a factor of two, $2^{N_{\max}} \leq L < 2^{N_{\max}+1}$, and with it R_Λ to a factor of two and ρ_Λ to a factor of four, from a laboratory count. If devices carry states spread across well over 205 logical qubits with no such wall, the account given here fails at its premise.

The framework also requires Λ to be constant, since L is. Reference [5] turns the lunar-laser-ranging bound on $\dot{G}/G$ [45] into $|1 + w_0| \lesssim 5 \times 10^{-4}$ for the present dark-energy equation of state. DESI DR2 prefers evolving dark energy over a constant Λ at 3.1σ with the cosmic microwave background, and at 2.8σ to 4.2σ with supernovae added, depending on the sample [46]; the Lyman- α analysis of the same release gives 2.7σ with the cosmic microwave background and 3.2σ with supernovae added [47]. A confirmed time variation of the dark-energy density would falsify the identification of R_Λ with a fixed length in Axiom 2; the framework predicts that the preference will not survive. For the same reason a_0 , tied to R_Λ and not to $H(z)$, does not evolve with redshift [5]. A rise of the characteristic acceleration of the radial acceleration relation to $2.4 \times 10^{-10} \text{ ms}^{-2}$ at $z \approx 1$ has been reported [48]; it implies, as its authors note, a shift of the baryonic Tully–Fisher zero point of -0.2 dex over the same interval, which the companion analysis, at two effective radii rather than in the flat part of the curve, does not find, 0.00 ± 0.06 dex [49]. The framework predicts that the rise will not survive.

VII. CONCLUSION

The vacuum catastrophe has two parts. The zero-point energy never enters the field equations, because in an emergent theory those equations are unchanged by a constant shift of the matter Lagrangian and Λ appears as a boundary condition. The scale of that boundary condition is set by the state count. The continuum assigns a horizon volume $(R/\ell_P)^3$ modes at E_P apiece and returns the Planck density for any R . A universe of finite information has L^2/π pixels on its horizon, each at the energy scale E_P/L , and the energy Λ carries inside the horizon is $S_{\text{BH}}T_{\text{dS}}$. Once the horizon is L Planck lengths around, $\rho_\Lambda/\rho_{\text{Planck}} = 3\pi/2L^2$ is an identity, the only number the counting can give, and once a quantum state cannot hold unlimited information, the same L bounds the state. The integer L has been determined from the horizon and from galactic rotation curves, and the two agree at 1.4σ of the galactic measurement, the residual being the coefficient

the framework has yet to derive. A third determination, with no astronomy in it, awaits the quantum computer: a ceiling of 205 usefully entangled qubits.

Appendix A: Numerical inputs

The inputs are $c = 2.99792458 \times 10^8 \text{ ms}^{-1}$, $\hbar = 1.0546 \times 10^{-34} \text{ Js}$, $G = 6.6743 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$, $\ell_P = 1.6163 \times 10^{-35} \text{ m}$, $E_P = \hbar c/\ell_P = 1.956 \times 10^9 \text{ J}$; $\Lambda = 1.09 \times 10^{-52} \text{ m}^{-2}$ from $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $\Omega_\Lambda = 0.685$ [37]; $a_0 = (1.20 \pm 0.02_{\text{stat}} \pm 0.24_{\text{sys}}) \times 10^{-10} \text{ ms}^{-2}$ [36]. Uncertainties are propagated linearly from the systematic term of a_0 ; the tighter $a_0 = (1.19 \pm 0.04_{\text{stat}} \pm 0.09_{\text{sys}}) \times 10^{-10} \text{ ms}^{-2}$ [44], its two terms added in quadrature, enters only the $\gamma = 1.38 \pm 0.11$ and the 3.3σ of Sec. V. Derived quantities are collected in Table I.

TABLE I. Derived quantities.

Quantity	Expression	Value
ρ_{Planck}	$c^7/\hbar G^2$	$4.63 \times 10^{113} \text{ J m}^{-3}$
ρ_Λ	$\Lambda c^4/8\pi G$	$5.25 \times 10^{-10} \text{ J m}^{-3}$
R_Λ	$\sqrt{3/\Lambda}$	$1.659 \times 10^{26} \text{ m}$
L	$2\pi R_\Lambda/\ell_P$	6.449×10^{61}
L_{gal}	$c^2/a_0\ell_P$	$(4.6 \pm 0.9) \times 10^{61}$
γ	$a_0 L \ell_P / c^2$	1.39 ± 0.28
N_{pix}	L^2/π	1.324×10^{123}
N_{cell}	L^2	4.159×10^{123}
S_{BH}/k_B	$L^2/4\pi$	3.310×10^{122}
Bulk modes	$(R_\Lambda/\ell_P)^3$	1.08×10^{183}
ε	E_P/L	$3.03 \times 10^{-53} \text{ J}$
$\rho_\Lambda/\rho_{\text{Planck}}$	$3\pi/2L^2$	1.133×10^{-123}
$\rho_\Lambda V$	$S_{\text{BH}}T_{\text{dS}}$	$1.004 \times 10^{70} \text{ J}$
$\rho_\Lambda^{1/4}$	$(3\pi/2)^{1/4} E_P/\sqrt{L}$	2.24 meV
$N_{\max}$	$\lfloor \log_2 L \rfloor$	205

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